

**Maulana Azad National Institute of Technology Bhopal Department  
of Mathematics, Bioinformatics and Computer Applications  
Mathematics-IV (MTH-241)**

**Assignment-6**

**Due Date:**

**Even Sem: 2024-25**

|    |                                                                                                                                                                                                                           |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Find the critical and stationary points of the following function.<br>$f(x) = \frac{x^3}{3} + y^2 + 2xy - 6x - 3y + 4$<br>Classify the stationary points into local maximum, local minimum, and saddle points.            |
| 2. | Find the critical and stationary points of the following function.<br>$f(x) = 4x^2 + 9y^2 + 8x - 36y + 24$<br>Classify the stationary points into local maximum, local minimum, and saddle points.                        |
| 3. | Find the dimensions of a rectangular parallelopiped with the largest volume whose sides are parallel to the co-ordinate planes to be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ |
| 4. | A positive quantity $b$ is to be divided into $n$ parts in such a way that the product of $n$ parts is to be a maximum. Use Lagrange's multiplier technique to obtain the optimal sub-division.                           |
| 5. | Optimize $Z = 50x - 2x^2 - xy - 3y^2 + 95y$<br>Sub. to : $x + y = 25$ ;<br>$x, y \geq 0$                                                                                                                                  |
| 6. | Optimize $2x + 3y + z$<br>Subject to: $x^2 + y^2 + z^2 = 1$<br>$x, y, z \geq 0$                                                                                                                                           |
| 7. | Optimize $f(x) = 4x_1 + 2x_2^2 + x_3^2 - 4x_1x_2$<br>Subject to: $x_1 + x_2 + x_3 = 15$<br>$2x_1 - x_2 + x_3 = 20$<br>$x_1, x_2, x_3 \geq 0$                                                                              |